

Bab_2_SAW_TOPSIS

December 22, 2025

```
[2]: import numpy as np
import pandas as pd
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[56]: config = {
    'C_1': {'bobot': 0.225, 'tipe': 'benefit'}, # Harus Padi
    'C_2': {'bobot': 0.225, 'tipe': 'benefit'}, # KRUSIAL: Harus SVM
    'C_3': {'bobot': 0.175, 'tipe': 'benefit'}, # KRUSIAL: Harus Warna+Tekstur
    'C_4': {'bobot': 0.175, 'tipe': 'benefit'}, # Utamakan data sawah/pribadi
    'C_5': {'bobot': 0.05, 'tipe': 'benefit'}, # Akurasi tinggi itu bonus
    'C_6': {'bobot': 0.15, 'tipe': 'benefit'} # Semakin baru semakin baik
}
```

No	Peneliti & Tahun	C_1 (Obj)	C_2 (Metd)	C_3 (Fitr)	C_4 (Cond)	C_5 (Acc)	C_6 (Thn)
1	Azmi et al.	2	1	1	0	0.99	2024
2	Sheila et al.	2	1	1	0	0.9375	2023
3	Oktaviana et al.	2	1	1	0	1	2021
4	Sethy et al.	2	3	1	1	0.9838	2020
5	Prajapati et al.	2	3	3	0	0.7333	2017
6	Ghyar & Birajdar	2	3	3	0	0.925	2017
7	Bashir et al.	2	3	1	0	0.9416	2019
8	Singh et al.	2	3	3	0	0.961	2022
10	Setiawan et al.	2	3	1	0	0.5381	2023
11	Rahman et al.	2	1	1	1	0.9433	2020
12	Susanto et al.	2	1	1	0	0.9647	2025

No	Peneliti & Tahun	C_1 (Obj)	C_2 (Metd)	C_3 (Fitr)	C_4 (Cond)	C_5 (Acc)	C_6 (Thn)
13	Fauzi et al.	2	1	1	0	0.932	2024
14	Agustiani et al.	2	2	2	0	0.9965	2022
15	Setiaji & Huda	2	2	2	0	0.933	2022
16	Purnamawati et al.	2	2	1	0	0.87	2020
17	Novantara et al.	2	1	1	1	0.9775	2025
18	Khoiruddin et al.	2	1	1	0	0.98	2022
19	Wagle & Harikrishnan	1	3	1	0	0.9115	2021
20	Burhanuddin	2	2	3	1	0.76	2024
21	Agustina et al.	2	1	1	0	0.9917	2024
23	Kusanti & Haris	2	2	2	1	0.80	2018
24	Rustanto et al.	2	2	2	0	0.9167	2024
25	Sarasati	2	1	1	1	0.9453	2021
26	Naresh kumar & Sakthivel	2	2	1	0	0.976	2025
28	Yao et al.	2	3	3	1	0.972	2009
29	Devi & Nee-lamegam	2	3	3	1	0.9863	2019
30	Hasan et al.	2	3	1	1	0.975	2019
31	Jiang et al.	2	3	1	1	0.968	2020
32	Verawati & Aunur-rohim	2	2	2	0	0.9167	2024
33	Milano	2	1	1	0	0.7705	2024
35	Saputra et al.	2	1	1	0	0.92	2021
36	Wenda et al.	2	2	3	0	0.907	2020
37	Hatuwal et al.	1	1	1	0	0.9789	2020

No	Peneliti & Tahun	C_1 (Obj)	C_2 (Metd)	C_3 (Fitr)	C_4 (Cond)	C_5 (Acc)	C_6 (Thn)
38	Pothen & Pai	2	3	2	0	0.946	2020
43	Das et al.	1	3	2	0	0.876	2020
44	Santosa et al.	2	1	1	0	0.9966	2023
45	Shah et al.	2	1	1	0	0.9975	2023
46	Istiqomah & Murinto	2	1	1	0	0.93	2024
47	Murniyasih & Suryani	2	2	1	1	0.75	2020
48	Hartono et al.	2	1	1	0	0.9375	2025

No, "Peneliti & Tahun", C_1, C_2, C_3, C_4, C_5, C_6 1, "Azmi et al.", 2, 1, 1, 0, 0.99, 2024 2, "Sheila et al.", 2, 1, 1, 0, 0.9375, 2023 3, "Oktaviana et al.", 2, 1, 1, 0, 1, 2021 4, "Sethy et al.", 2, 3, 1, 1, 0.9838, 2020 5, "Prajapati et al.", 2, 3, 3, 0, 0.7333, 2017 6, "Ghyar & Birajdar", 2, 3, 3, 0, 0.925, 2017 7, "Bashir et al.", 2, 3, 1, 0, 0.9416, 2019 8, "Singh et al.", 2, 3, 3, 0, 0.961, 2022 10, "Setiawan et al.", 2, 3, 1, 0, 0.5381, 2023 11, "Rahman et al.", 2, 1, 1, 1, 0.9433, 2020 12, "Susanto et al.", 2, 1, 1, 0, 0.9647, 2025 13, "Fauzi et al.", 2, 1, 1, 0, 0.932, 2024 14, "Agustiani et al.", 2, 2, 2, 0, 0.9965, 2022 15, "Setiaji & Huda", 2, 2, 2, 0, 0.933, 2022 16, "Purnamawati et al.", 2, 2, 1, 0, 0.87, 2020 17, "Novantara et al.", 2, 1, 1, 1, 0.9775, 2025 18, "Khoiruddin et al.", 2, 1, 1, 0, 0.98, 2022 19, "Wagle & Harikrishnan", 1, 3, 1, 0, 0.9115, 2021 20, "Burhanuddin", 2, 2, 3, 1, 0.76, 2024 21, "Agustina et al.", 2, 1, 1, 0, 0.9917, 2024 23, "Kusanti & Haris", 2, 2, 2, 1, 0.80, 2018 24, "Rustanto et al.", 2, 2, 2, 0, 0.9167, 2024 25, "Sarasati", 2, 1, 1, 1, 0.9453, 2021 26, "Naresh kumar & Sakthivel", 2, 2, 1, 0, 0.976, 2025 28, "Yao et al.", 2, 3, 3, 1, 0.972, 2009 29, "Devi & Neelamegam", 2, 3, 3, 1, 0.9863, 2019 30, "Hasan et al.", 2, 3, 1, 1, 0.975, 2019 31, "Jiang et al.", 2, 3, 1, 1, 0.968, 2020 32, "Verawati & Aunurrohim", 2, 2, 2, 0, 0.9167, 2024 33, "Milano", 2, 1, 1, 0, 0.7705, 2024 35, "Saputra et al.", 2, 1, 1, 0, 0.92, 2021 36, "Wenda et al.", 2, 2, 3, 0, 0.907, 2020 37, "Hatuwal et al.", 1, 1, 1, 0, 0.9789, 2020 38, "Pothen & Pai", 2, 3, 2, 0, 0.946, 2020 43, "Das et al.", 1, 3, 2, 0, 0.876, 2020 44, "Santosa et al.", 2, 1, 1, 0, 0.9966, 2023 45, "Shah et al.", 2, 1, 1, 0, 0.9975, 2023 46, "Istiqomah & Murinto", 2, 1, 1, 0, 0.93, 2024 47, "Murniyasih & Suryani", 2, 2, 1, 1, 0.75, 2020 48, "Hartono et al.", 2, 1, 1, 0, 0.9375, 2025

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[32]: df_data = pd.read_csv('seleksi bab 2.txt', index_col=0)
df_data.sort_values('C_6')
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[32]:
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No	Peneliti & Tahun	C_1	C_2	C_3	C_4	C_5	C_6
28	Yao et al.	2	3	3	1	0.9720	2009
6	Ghyar & Birajdar	2	3	3	0	0.9250	2017
5	Prajapati et al.	2	3	3	0	0.7333	2017

23	Kusanti & Haris	2	2	2	1	0.8000	2018
30	Hasan et al.	2	3	1	1	0.9750	2019
29	Devi & Neelamegam	2	3	3	1	0.9863	2019
7	Bashir et al.	2	3	1	0	0.9416	2019
4	Sethy et al.	2	3	1	1	0.9838	2020
16	Purnamawati et al.	2	2	1	0	0.8700	2020
11	Rahman et al.	2	1	1	1	0.9433	2020
36	Wenda et al.	2	2	3	0	0.9070	2020
31	Jiang et al.	2	3	1	1	0.9680	2020
47	Murniyasih & Suryani	2	2	1	1	0.7500	2020
43	Das et al.	1	3	2	0	0.8760	2020
38	Pothen & Pai	2	3	2	0	0.9460	2020
37	Hatuwal et al.	1	1	1	0	0.9789	2020
19	Wagle & Harikrishnan	1	3	1	0	0.9115	2021
25	Sarasati	2	1	1	1	0.9453	2021
3	Oktaviana et al.	2	1	1	0	1.0000	2021
35	Saputra et al.	2	1	1	0	0.9200	2021
8	Singh et al.	2	3	3	0	0.9610	2022
18	Khoiruddin et al.	2	1	1	0	0.9800	2022
15	Setiaji & Huda	2	2	2	0	0.9330	2022
14	Agustiani et al.	2	2	2	0	0.9965	2022
2	Sheila et al.	2	1	1	0	0.9375	2023
10	Setiawan et al.	2	3	1	0	0.5381	2023
45	Shah et al.	2	1	1	0	0.9975	2023
44	Santosa et al.	2	1	1	0	0.9966	2023
33	Milano	2	1	1	0	0.7705	2024
24	Rustanto et al.	2	2	2	0	0.9167	2024
13	Fauzi et al.	2	1	1	0	0.9320	2024
1	Azmi et al.	2	1	1	0	0.9900	2024
21	Agustina et al.	2	1	1	0	0.9917	2024
20	Burhanuddin	2	2	3	1	0.7600	2024
46	Istiqomah & Murinto	2	1	1	0	0.9300	2024
32	Verawati & Aunurrohim	2	2	2	0	0.9167	2024
26	Naresh kumar & Sakthivel	2	2	1	0	0.9760	2025
17	Novantara et al.	2	1	1	1	0.9775	2025
12	Susanto et al.	2	1	1	0	0.9647	2025
48	Hartono et al.	2	1	1	0	0.9375	2025

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[33]: df_C_only = df_data.drop(['Peneliti & Tahun'], axis=1)
df_C_only.head()
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[33]:
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	C_1	C_2	C_3	C_4	C_5	C_6
No						
1	2	1	1	0	0.9900	2024
2	2	1	1	0	0.9375	2023
3	2	1	1	0	1.0000	2021
4	2	3	1	1	0.9838	2020

```
5      2      3      3      0  0.7333  2017
```

```
[57]: df_calc = df_C_only.copy()
df_norm = pd.DataFrame()
kriteria_cols = list(config.keys())

for col, settings in config.items():
    if settings['tipe']=='benefit':
        df_norm[col] = df_calc[col]/df_calc[col].max()
    elif settings['tipe']=='cost':
        df_norm[col] = df_calc[col].min()/df_calc[col]

df_weighted = df_norm.copy()
for col, settings in config.items():
    df_weighted[col] = df_norm[col] * settings['bobot']

ideal_positif = {}
ideal_negatif = {}

for col in kriteria_cols:
    # Karena sudah dinormalisasi SAW (Benefit semua arahnya positif),
    # maka Ideal Positif selalu Max, Negatif selalu Min
    ideal_positif[col] = df_weighted[col].max()
    ideal_negatif[col] = df_weighted[col].min()

def get_distance(row, ideal_dict):
    dist = 0
    for col in kriteria_cols:
        dist += (row[col] - ideal_dict[col]) ** 2
    return np.sqrt(dist)

df_calc['D_plus'] = df_weighted.apply(lambda x: get_distance(x, ideal_positif),
    ↪axis=1)
df_calc['D_min'] = df_weighted.apply(lambda x: get_distance(x, ideal_negatif),
    ↪axis=1)

df_calc['Skor_Akhir'] = df_calc['D_min'] / (df_calc['D_min'] +
    ↪df_calc['D_plus'])

df_calc.sort_values(by='Skor_Akhir', ascending=False)
```

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[57]:
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	C_1	C_2	C_3	C_4	C_5	C_6	D_plus	D_min	Skor_Akhir
No									
29	2	3	3	1	0.9863	2019	0.000817	0.282657	0.997119
28	2	3	3	1	0.9720	2009	0.001834	0.282600	0.993551
20	2	2	3	1	0.7600	2024	0.075954	0.250283	0.767182
23	2	2	2	1	0.8000	2018	0.095541	0.229087	0.705691

4	2	3	1	1	0.9838	2020	0.116670	0.257446	0.688145
30	2	3	1	1	0.9750	2019	0.116674	0.257408	0.688106
31	2	3	1	1	0.9680	2020	0.116678	0.257379	0.688074
47	2	2	1	1	0.7500	2020	0.139257	0.221403	0.613883
8	2	3	3	0	0.9610	2022	0.175011	0.221845	0.559007
6	2	3	3	0	0.9250	2017	0.175041	0.221680	0.558780
5	2	3	3	0	0.7333	2017	0.175508	0.221050	0.557421
17	2	1	1	1	0.9775	2025	0.190033	0.209202	0.524007
25	2	1	1	1	0.9453	2021	0.190049	0.209037	0.523789
11	2	1	1	1	0.9433	2020	0.190051	0.209027	0.523775
38	2	3	2	0	0.9460	2020	0.184486	0.197423	0.516936
36	2	2	3	0	0.9070	2020	0.190451	0.179536	0.485249
7	2	3	1	0	0.9416	2019	0.210345	0.188584	0.472726
10	2	3	1	0	0.5381	2023	0.211588	0.187503	0.469825
43	1	3	2	0	0.8760	2020	0.216154	0.161830	0.428139
14	2	2	2	0	0.9965	2022	0.199130	0.149031	0.428052
15	2	2	2	0	0.9330	2022	0.199158	0.148576	0.427269
32	2	2	2	0	0.9167	2024	0.199174	0.148471	0.427077
24	2	2	2	0	0.9167	2024	0.199174	0.148471	0.427077
19	1	3	1	0	0.9115	2021	0.238562	0.151160	0.387866
26	2	2	1	0	0.9760	2025	0.223299	0.136975	0.380196
16	2	2	1	0	0.8700	2020	0.223391	0.136225	0.378807
3	2	1	1	0	1.0000	2021	0.258334	0.114850	0.307757
45	2	1	1	0	0.9975	2023	0.258333	0.114826	0.307712
44	2	1	1	0	0.9966	2023	0.258333	0.114817	0.307696
21	2	1	1	0	0.9917	2024	0.258334	0.114769	0.307607
1	2	1	1	0	0.9900	2024	0.258334	0.114752	0.307575
18	2	1	1	0	0.9800	2022	0.258335	0.114653	0.307391
12	2	1	1	0	0.9647	2025	0.258339	0.114510	0.307122
48	2	1	1	0	0.9375	2025	0.258352	0.114265	0.306655
2	2	1	1	0	0.9375	2023	0.258352	0.114263	0.306652
13	2	1	1	0	0.9320	2024	0.258356	0.114216	0.306562
46	2	1	1	0	0.9300	2024	0.258357	0.114199	0.306529
35	2	1	1	0	0.9200	2021	0.258364	0.114112	0.306361
33	2	1	1	0	0.7705	2024	0.258588	0.113104	0.304295
37	1	1	1	0	0.9789	2020	0.281769	0.022055	0.072592